Data Transformation II

PSY 410: Data Science for Psychology

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From rows to statistics

Your advisor's first question

Last time, you learned to pick rows, sort them, and create new columns.

But your advisor doesn't want rows. They want a number:

"What's the average depression score by condition?"

Today we learn to answer that — with **group_by()** + **summarize()**.

summarize()

summarize() collapses data

Reduces a data frame to a single row of summary statistics:

```
flights |>
  summarize(
    avg_delay = mean(dep_delay, na.rm = TRUE),
    max_delay = max(dep_delay, na.rm = TRUE),
    n_flights = n()
  )
```

```
# A tibble: 1 × 3
  avg_delay max_delay n_flights
      <dbl>     <dbl>     <int>
1      12.6      1301    336776
```

Key summary functions

Function	What it computes
`mean()`, `median()`	Central tendency
`sd()`, `var()`	Spread
`min()`, `max()`	Extremes
`sum()`	Total
`n()`	Count of rows
`n_distinct()`	Count of unique values

The na.rm argument

Most summary functions need `na.rm = TRUE` to handle missing values:

```
x <- c(1, 2, NA, 4, 5)

mean(x)                    # Returns NA
```

```
[1] NA
```

```
mean(x, na.rm = TRUE)      # Returns 3
```

```
[1] 3
```

Tip

Always use `na.rm = TRUE` unless you specifically want NA propagation.

summarize() alone isn't very useful

One row of summary stats? That's what `mean()` already does.

```
mean(flights$dep_delay, na.rm = TRUE)
```

```
[1] 12.63907
```

The power comes when we combine it with **group_by()**.

group_by()

group_by() + summarize() = magic

```r
flights |>
  group_by(month) |>
  summarize(
    avg_delay = mean(dep_delay, na.rm = TRUE),
    n_flights = n()
  )
```

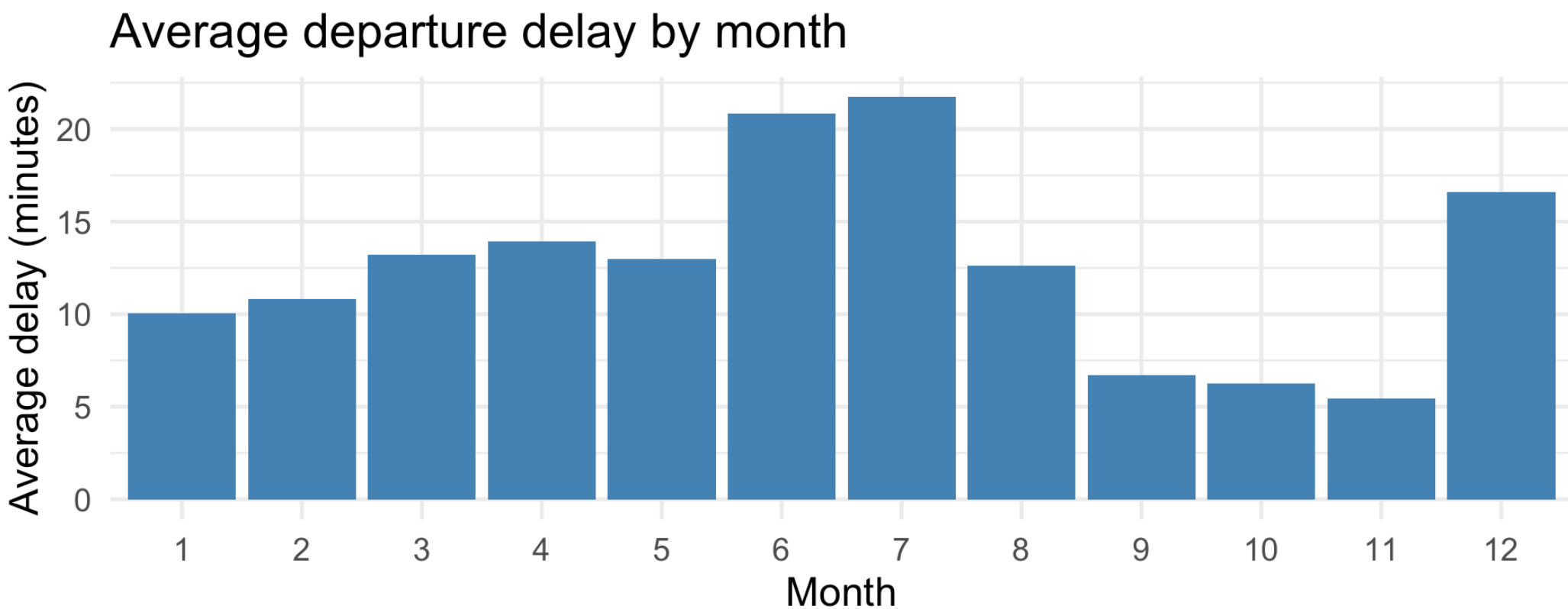
```
# A tibble: 12 × 3
   month avg_delay n_flights
   <int>     <dbl>     <int>
 1     1     10.0      27004
 2     2     10.8      24951
 3     3     13.2      28834
 4     4     13.9      28330
 5     5     13.0      28796
 6     6     20.8      28243
 7     7     21.7      29425
 8     8     12.6      29327
 9     9      6.72     27574
```

```
10     10       6.24      28889
11     11       5.44      27268
```

What happened?

1. `group_by(month)` split the data into 12 groups (one per month)
2. `summarize()` calculated statistics **within each group**
3. Result: one row per group

Visualizing group_by()



Group by multiple variables

```r
# Average delay by month AND carrier
flights |>
  group_by(month, carrier) |>
  summarize(
    avg_delay = mean(dep_delay, na.rm = TRUE),
    n_flights = n()
  )
```

```
# A tibble: 185 × 4
# Groups:   month [12]
   month carrier avg_delay n_flights
   <int> <chr>       <dbl>     <int>
 1     1 9E          16.9       1573
 2     1 AA           6.93      2794
 3     1 AS           7.35        62
 4     1 B6           9.49      4427
 5     1 DL           3.85      3690
 6     1 EV          24.2       4171
 7     1 F9          10           59
```

```
 8      1 FL            1.97        328
 9      1 HA           54.4          31
10      1 MQ            6.49       2271
```

The .groups argument

You may see this message:

```
`summarise()` has grouped output by 'month'. You can override using the
`.groups` argument.
```

Control this with:

```r
# Keep grouping (default with multiple groups)
summarize(..., .groups = "keep")

# Drop all grouping
summarize(..., .groups = "drop")

# Drop last group level
summarize(..., .groups = "drop_last")
```

Multiple summaries at once

```r
flights |>
  group_by(carrier) |>
  summarize(
    # Central tendency
    mean_delay = mean(arr_delay, na.rm = TRUE),
    median_delay = median(arr_delay, na.rm = TRUE),
    # Spread
    sd_delay = sd(arr_delay, na.rm = TRUE),
    # Counts
    n_flights = n(),
    n_dest = n_distinct(dest)
  ) |>
  arrange(mean_delay)  # Best to worst carriers
```

Multiple summaries at once

```
# A tibble: 16 × 6
   carrier mean_delay median_delay sd_delay n_flights n_dest
   <chr>        <dbl>        <dbl>    <dbl>     <int>  <int>
 1 AS          -9.93           -17     36.5       714      1
 2 HA          -6.92           -13     75.1       342      1
 3 AA           0.364           -9     42.5     32729     19
 4 DL           1.64            -8     44.4     48110     40
 5 VX           1.76            -9     50.0      5162      5
 6 US           2.13            -6     33.1     20536      6
 7 UA           3.56            -6     41.0     58665     47
 8 9E           7.38            -7     50.1     18460     49
 9 B6           9.46            -3     42.8     54635     42
10 WN           9.65            -3     46.9     12275     11
11 MQ          10.8             -1     43.2     26397     20
12 OO          11.9              7     48.6        32      5
```

Pair coding break

Your turn: 10 minutes

With a partner, using the `flights` dataset:

1. Calculate the **average departure delay** for each carrier
2. Which airline has the **worst** average delay?
3. **Bonus:** Also calculate the number of flights per carrier. Does the worst airline just have fewer flights?

Before we move on

📤 Submit your code on Canvas for participation credit. Paste what you have — it doesn't need to work perfectly.

What you can already do

With `summarize()` + `group_by()`, you can answer questions like:

- *"What's the average depression score by condition?"*
- *"Which group had the most variability?"*
- *"How many participants completed each phase?"*

These are the building blocks of a results section. Next: some shortcuts.

count() — a shortcut

Counting is common

```r
# How many flights per carrier?
flights |>
  group_by(carrier) |>
  summarize(n = n())
```

```
# A tibble: 16 × 2
   carrier     n
   <chr>   <int>
 1 9E      18460
 2 AA      32729
 3 AS        714
 4 B6      54635
 5 DL      48110
 6 EV      54173
 7 F9        685
 8 FL       3260
 9 HA        342
10 MQ      26397
11 OO         32
12 UA      58665
```

count() does this automatically

```
flights |>
  count(carrier)
```

```
# A tibble: 16 × 2
   carrier     n
   <chr>   <int>
 1 9E      18460
 2 AA      32729
 3 AS        714
 4 B6      54635
 5 DL      48110
 6 EV      54173
 7 F9        685
 8 FL       3260
 9 HA        342
10 MQ      26397
11 OO         32
12 UA      58665
```

Same result, less typing!

count() with multiple variables

```r
flights |>
  count(carrier, origin)
```

```
# A tibble: 35 × 3
   carrier origin     n
   <chr>   <chr>  <int>
 1 9E      EWR     1268
 2 9E      JFK    14651
 3 9E      LGA     2541
 4 AA      EWR     3487
 5 AA      JFK    13783
 6 AA      LGA    15459
 7 AS      EWR      714
 8 B6      EWR     6557
 9 B6      JFK    42076
10 B6      LGA     6002
# ℹ 25 more rows
```

Adding weights

```r
# Total distance flown by each carrier
flights |>
  count(carrier, wt = distance, name = "total_miles")
```

```
# A tibble: 16 × 2
   carrier total_miles
   <chr>         <dbl>
 1 9E          9788152
 2 AA         43864584
 3 AS          1715028
 4 B6         58384137
 5 DL         59507317
 6 EV         30498951
 7 F9          1109700
 8 FL          2167344
 9 HA          1704186
10 MQ         15033955
11 OO            16026
12 UA         89705524
```

Combining operations

The full dplyr toolkit

Now you can combine everything:

```
data |>
  filter()     |>   # Subset rows
  select()     |>   # Pick columns
  mutate()     |>   # Create columns
  group_by()   |>   # Define groups
  summarize()  |>   # Calculate summaries
  arrange()          # Sort results
```

A complete analysis pipeline

```r
# Which airlines were most delayed in summer?
flights |>
  filter(month %in% c(6, 7, 8)) |>    # Summer months only
  filter(!is.na(arr_delay)) |>         # Remove missing
  group_by(carrier) |>                 # Group by airline
  summarize(
    avg_delay = mean(arr_delay),
    pct_delayed = mean(arr_delay > 0) * 100,  # % of flights delayed
    n_flights = n()
  ) |>
  filter(n_flights > 1000) |>          # Only airlines with many flights
  arrange(desc(avg_delay))             # Worst first
```

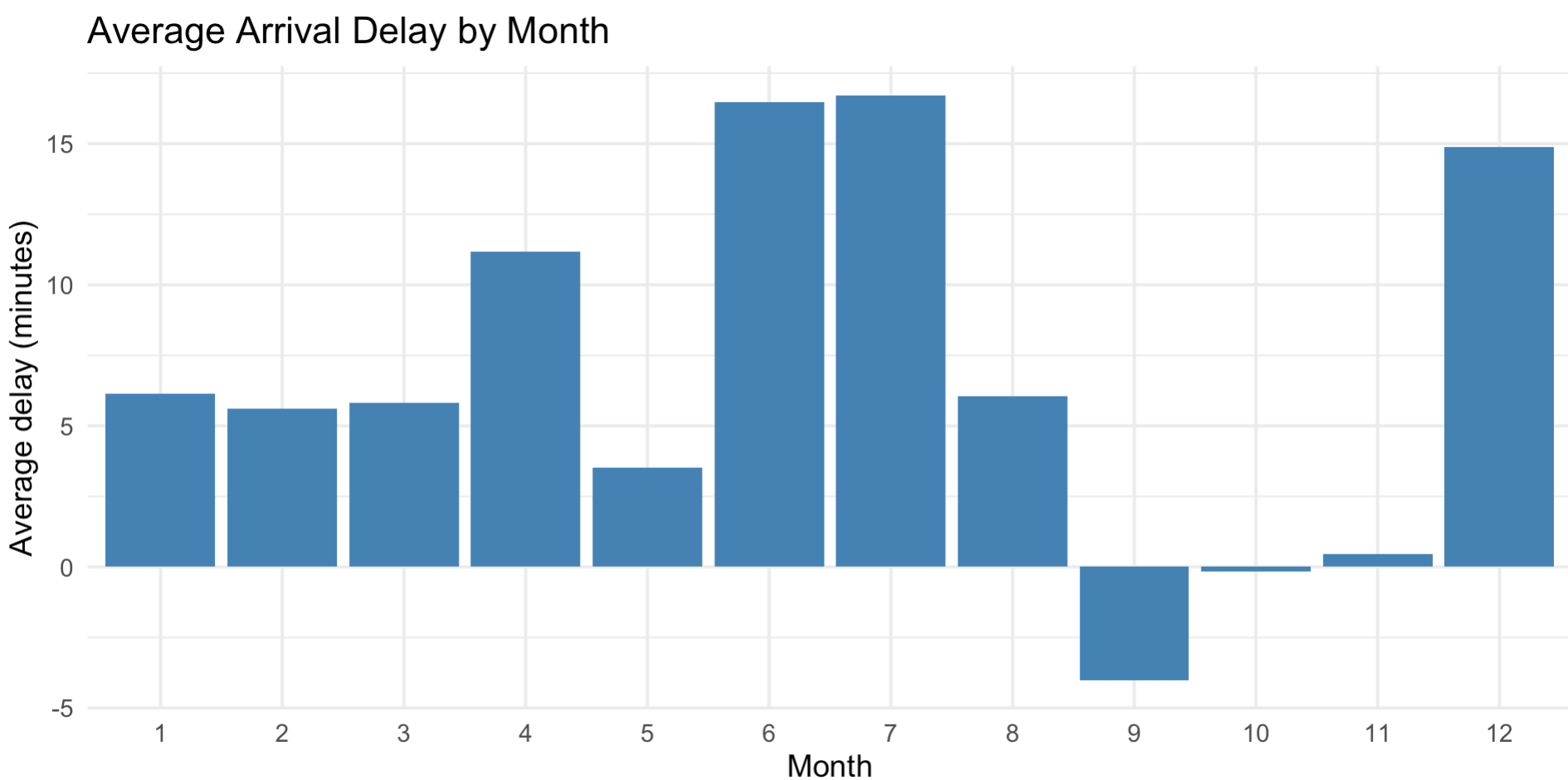
A complete analysis pipeline

```
# A tibble: 10 × 4
   carrier avg_delay pct_delayed n_flights
   <chr>       <dbl>       <dbl>     <int>
 1 MQ          18.6         52.6      6254
 2 EV          17.9         47.2     12715
 3 B6          17.7         48.9     14393
 4 9E          17.0         44.9      3972
 5 WN          16.6         47.2      3111
 6 VX          15.9         42.3      1451
 7 DL           9.56        40.9     12568
 8 UA           8.91        42.1     14941
 9 US           7.90        42.4      5086
10 AA           2.76        34.1      8271
```

Piping into ggplot

```r
flights |>
  filter(!is.na(arr_delay)) |>
  group_by(month) |>
  summarize(avg_delay = mean(arr_delay)) |>
  ggplot(aes(x = factor(month), y = avg_delay)) +
  geom_col(fill = "steelblue") +
  labs(
    title = "Average Arrival Delay by Month",
    x = "Month",
    y = "Average delay (minutes)"
  ) +
  theme_minimal(base_size = 14)
```

Piping into ggplot



group_by() with mutate()

`group_by()` also affects `mutate()` and `filter()`:

```
# Add a column for mean delay PER CARRIER
flights |>
  group_by(carrier) |>
  mutate(
    carrier_mean_delay = mean(arr_delay, na.rm = TRUE),
    delay_vs_carrier = arr_delay - carrier_mean_delay
  ) |>
  select(carrier, arr_delay, carrier_mean_delay, delay_vs_carrier)
```

group_by() with mutate()

`group_by()` also affects `mutate()` and `filter()`:

```
# A tibble: 336,776 × 4
# Groups:   carrier [16]
   carrier arr_delay carrier_mean_delay delay_vs_carrier
   <chr>       <dbl>              <dbl>            <dbl>
 1 UA             11               3.56             7.44
 2 UA             20               3.56            16.4
 3 AA             33               0.364           32.6
 4 B6            -18               9.46           -27.5
 5 DL            -25               1.64           -26.6
 6 UA             12               3.56             8.44
 7 B6             19               9.46             9.54
 8 EV            -14              15.8            -29.8
 9 B6             -8               9.46           -17.5
10 AA              8               0.364            7.64
# ℹ 336,766 more rows
```

Filtering within groups

```r
# Find the 2 most delayed flights per month
flights |>
  group_by(month) |>
  filter(!is.na(arr_delay)) |>
  slice_max(arr_delay, n = 2) |>
  select(month, carrier, arr_delay)
```

```
# A tibble: 24 × 3
# Groups:   month [12]
   month carrier arr_delay
   <int> <chr>       <dbl>
 1     1 HA           1272
 2     1 MQ           1109
 3     2 F9            834
 4     2 DL            773
 5     3 DL            915
 6     3 DL            784
 7     4 DL            931
 8     4 DL            821
```

```
 9      5 MQ              875
10      5 AA              852
```

slice() variants

Function	What it does
`slice_head(n)`	First n rows
`slice_tail(n)`	Last n rows
`slice_max(var, n)`	n rows with highest var
`slice_min(var, n)`	n rows with lowest var
`slice_sample(n)`	Random n rows

All work within groups when preceded by `group_by()`.

ungroup()

Sometimes you need to remove grouping:

```
flights |>
  group_by(carrier) |>
  mutate(carrier_mean = mean(arr_delay, na.rm = TRUE)) |>
  ungroup() |>                                  # Remove grouping
  mutate(grand_mean = mean(arr_delay, na.rm = TRUE)) |>  # Now uses ALL fligh
  select(carrier, arr_delay, carrier_mean, grand_mean)
```

```
# A tibble: 336,776 × 4
   carrier arr_delay carrier_mean grand_mean
   <chr>       <dbl>        <dbl>      <dbl>
 1 UA             11        3.56        6.90
 2 UA             20        3.56        6.90
 3 AA             33        0.364       6.90
 4 B6            -18        9.46        6.90
 5 DL            -25        1.64        6.90
 6 UA             12        3.56        6.90
 7 B6             19        9.46        6.90
```

```
 8 EV              -14        15.8          6.90
 9 B6               -8         9.46         6.90
10 AA                8         0.364        6.90
# ℹ 336,766 more rows
```

Code style matters

Spot the difference

```
# Version A
pD<-read_csv("participants.csv")
m<-pD|>filter(age>25)|>group_by(condition)|>summarize(m=mean(score,na.rm=TRUE
```

```
# Version B
participant_data <- read_csv("participants.csv")

# Adults only — age cutoff matches preregistration
participant_data |>
  filter(age > 25) |>
  group_by(condition) |>
  summarize(
    mean_score = mean(score, na.rm = TRUE),
    sd_score = sd(score, na.rm = TRUE),
    n = n()
  )
```

Same code. One you can debug at midnight. One you can't.

The rules behind the difference

Spacing

- Spaces around `<-`, `==`, `|>`
- Spaces after commas
- One verb per line

Naming

- `snake_case`: `mean_score`, not `m`
- Meaningful: `participant_data`, not `pD`
- Comments explain **why**, not what

Practice: Psychology example

Simulated experiment data

```r
set.seed(42)

# Create a simulated experiment dataset
experiment <- tibble(
  participant_id = rep(1:60, each = 20),
  condition = rep(rep(c("control", "treatment"), each = 10), 60),
  trial = rep(1:20, 60),
  reaction_time = rnorm(1200, mean = 500, sd = 100) +
                  ifelse(rep(rep(c(0, 1), each = 10), 60) == 1, -30, 0),
  accuracy = rbinom(1200, 1, prob = 0.85 +
                    ifelse(rep(rep(c(0, 1), each = 10), 60) == 1, 0.05, 0))
)
```

Examine the data

```
experiment
```

```
# A tibble: 1,200 × 5
   participant_id condition trial reaction_time accuracy
            <int> <chr>     <int>         <dbl>    <int>
 1              1 control       1          637.        1
 2              1 control       2          444.        1
 3              1 control       3          536.        1
 4              1 control       4          563.        1
 5              1 control       5          540.        1
 6              1 control       6          489.        0
 7              1 control       7          651.        1
 8              1 control       8          491.        1
 9              1 control       9          702.        1
10              1 control      10          494.        1
# ℹ 1,190 more rows
```

Participant-level summaries

```r
# First, summarize BY PARTICIPANT
participant_summaries <- experiment |>
  group_by(participant_id, condition) |>
  summarize(
    mean_rt = mean(reaction_time),
    accuracy = mean(accuracy) * 100,
    n_trials = n(),
    .groups = "drop"
  )

participant_summaries
```

Participant-level summaries

```
# A tibble: 120 × 5
   participant_id condition mean_rt accuracy n_trials
            <int> <chr>       <dbl>    <dbl>    <int>
 1              1 control      555.       90       10
 2              1 treatment    454.       90       10
 3              2 control      482.       80       10
 4              2 treatment    434.       70       10
 5              3 control      498.       80       10
 6              3 treatment    472.      100       10
 7              4 control      554.       90       10
 8              4 treatment    448.       90       10
 9              5 control      525.      100       10
10              5 treatment    461.       80       10
# ℹ 110 more rows
```

Condition-level summaries

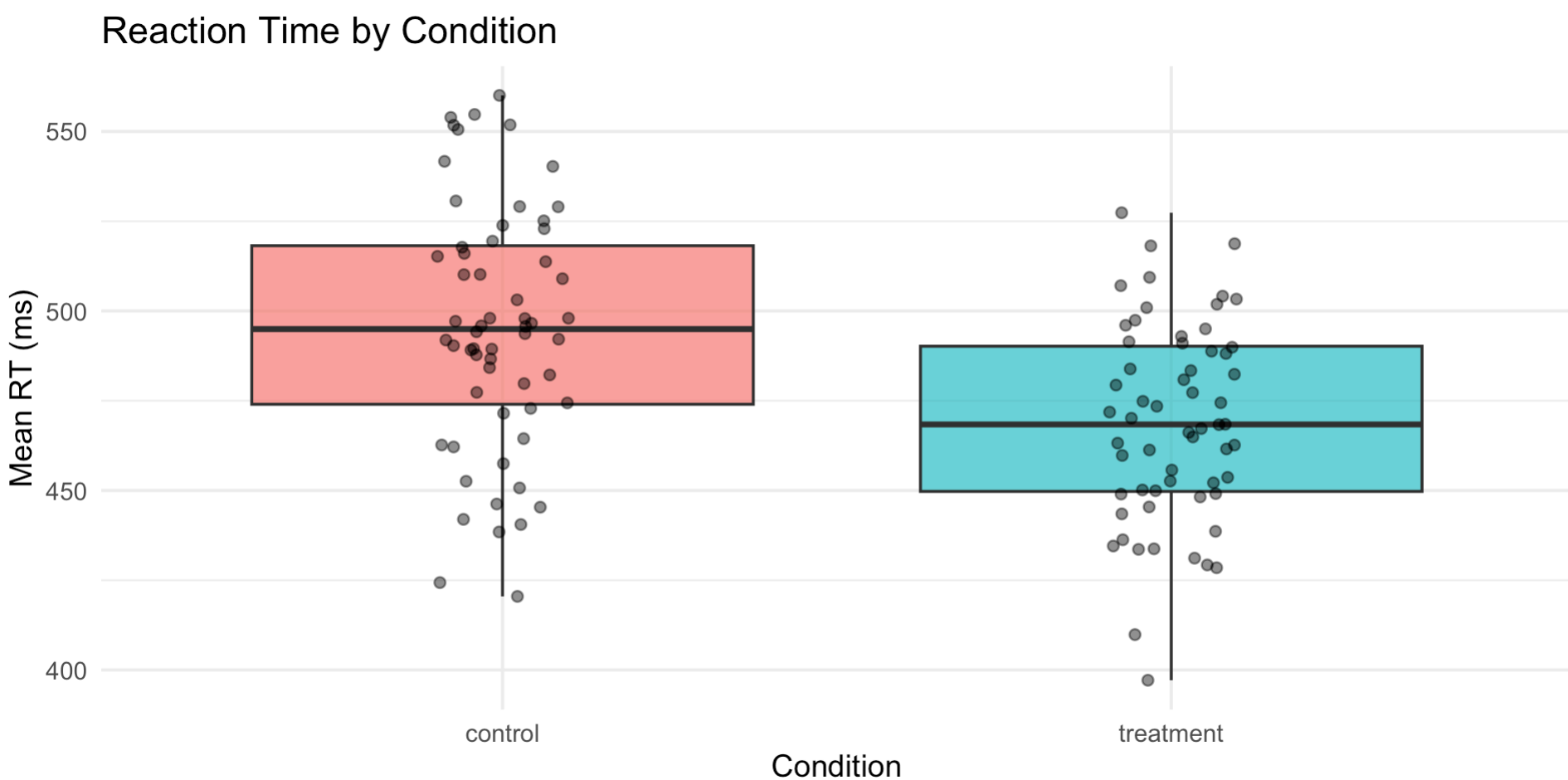
```r
# Now summarize ACROSS PARTICIPANTS by condition
condition_summaries <- participant_summaries |>
  group_by(condition) |>
  summarize(
    M_rt = mean(mean_rt),
    SD_rt = sd(mean_rt),
    M_acc = mean(accuracy),
    SD_acc = sd(accuracy),
    n = n()
  )

condition_summaries
```

```
# A tibble: 2 × 6
  condition  M_rt SD_rt M_acc SD_acc     n
  <chr>     <dbl> <dbl> <dbl>  <dbl> <int>
1 control    495.  34.3  84.8   11.4    60
2 treatment  469.  27.7  89.2   10.1    60
```

Visualize the results

```r
participant_summaries |>
  ggplot(aes(x = condition, y = mean_rt, fill = condition)) +
  geom_boxplot(alpha = 0.7) +
  geom_jitter(width = 0.1, alpha = 0.5) +
  labs(
    title = "Reaction Time by Condition",
    x = "Condition",
    y = "Mean RT (ms)"
  ) +
  theme_minimal(base_size = 14) +
  theme(legend.position = "none")
```

Visualize the results



Get a head start: Assignment 2

Using the `experiment` data we just created:

1. Filter to only accurate trials (`accuracy == 1`)
2. Calculate mean RT **per participant** per condition
3. Find the 5 participants with the **fastest** mean RT in each condition

> **Tip**
>
> This uses everything from today — `filter()`, `group_by()`, `summarize()`, and `slice_min()`. Work through it step by step.

Wrapping up

Today's additions

Verb	What it does
`group_by()`	Define groups for operations
`summarize()`	Calculate summary statistics
`count()`	Quick frequency counts
`ungroup()`	Remove grouping
`slice_*()`	Select rows by position/rank

The analysis workflow

```r
raw_data |>
  filter(valid_data) |>           # Clean
  select(relevant_vars) |>        # Focus
  mutate(new_vars) |>             # Transform
  group_by(conditions) |>         # Split
  summarize(statistics) |>        # Aggregate
  arrange(order) |>               # Sort
  ggplot() + ...                  # Visualize
```

Before next class

📖 **Read:**

- R4DS Ch 5: Data tidying

✅ **Practice:**

- Calculate descriptive statistics by group
- Build complete analysis pipelines
- Focus on writing clean, readable code

Key takeaways

1. **group_by() + summarize()** is incredibly powerful
2. **count()** is a quick shortcut for frequencies
3. **Grouped operations** work with mutate() and filter() too
4. **ungroup()** when you need to work with all data again
5. **Code style matters** — write for your future self

The one thing to remember

`group_by()` + `summarize()` is how you go from raw data to the descriptive statistics table in every published paper.

Next time: Data Tidying with tidyr